

Predicting household sign up for solar energy: an empirical study based on the extended theory of planned behavior

Extended
theory of
planned
behavior

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Abstract

Purpose – This study aims to understand households' adoption of small-scale solar energy to reduce carbon dioxide emissions that cause due to conventional energy consumptions.

Design/methodology/approach – This study is quantitative in nature and households were selected as unit of analysis. Online data has been collected from seven main cities of Pakistan to understand households' intention to use small-scale solar energy for domestic consumption. A total of 370 valid data were analyzed through partial least square structural equation modeling.

Findings – The study findings reveal that publicity information, attitude green norm and perceived behavioral control are the strongest predictors of households' intention to use small-scale solar energy.

Practical implications – The considered model practically contributes to the literature by understanding households' intention to adopt solar technologies that are viable means to conserve conventional energy and preserve the environment through less emission of carbon dioxide. In addition to this, understanding the green norm of households is imperative in a developing country, Pakistan where climate risk is high. Understanding household' green norms would help marketers and practitioners to design and introduce new and more efficient renewable technologies that maintain environmental sustainability.

Originality/value – This study has contributed to theory of planned behavior (TPB) by the inclusion of publicity information and green norms. Previous studies focused on the environmental benefits of using

Consent to participate.

Consent was obtained from the household prior to data collection.

Consent to publish.

The authors have agreed to publish the content in the journal.

Competing interests.

The authors declare that they have no competing interest.

Availability of the data and material.

The data sets used/or analyzed during the current study are available from the corresponding author on reasonable request.



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renewable energy sources. This study added novel antecedents into TPB that help to understand the adoption of small-scale solar energy for domestic consumption.

Keywords Publicity information, Green norms, Attitude, Perceived behavioral control, Subjective norms, Intention

Paper type Research paper

1. Introduction

Nature is experiencing environmental changes and biodiversity losses due to rapid climate change, ozone layer depletion and greenhouse gas emissions (UN, 2015; IPCC, 2011). Environmental scientists are positing that the human race is experiencing unusual rapid global warming (Waris and Hameed, 2020; Iqbal *et al.*, 2020). The emission of carbon dioxide (CO₂) into the environment has been recognized as one of the main causes of global warming (Elmustapha *et al.*, 2018). Due to the adverse effects of global warming, many countries have legislated to reduce the harmful effects of climate change (Ben Amara and Chen, 2020). The Paris agreement on climate change has proposed some strict measures and urged all nations to reduce the consumption of energy (Paris, 2015). These policies attracted the policymakers, economic and political scientists to formulate the strategies that could counter the challenges of global warming faced by society (Shepherd and Patzelt, 2011).

Literature depicts that energy consumption by domestic consumers has significantly contributed to the emission of carbon dioxide into the environment (Tang *et al.*, 2020; Renukappa *et al.*, 2013). Therefore, policy intervention regarding the use of renewable technologies is the need of time due to its capacity to reduce carbon dioxide emissions (Ebrahimi *et al.*, 2021; Elmustapha *et al.*, 2018). To reduce the dependency on fossil fuels, various countries have initiated policies to increase the production of solar energy at the domestic level (Siepermann *et al.*, 2021; Khanmohammadi *et al.*, 2021). Further, the developments in the energy sector have been amplified by the wave of digitalization (Ringel, 2018). However, the adoption of renewable technologies in developing countries is slow due people's unawareness and financial constraints (Kalair *et al.*, 2020; Khatib *et al.*, 2011).

The oil crisis of the 1970s has attracted the attention of consumers towards energy-related issues (Hartmann and Apaolaza-Ibañez, 2012; McDougall *et al.*, 1981). Past studies depict that consumers' awareness related to environmental issues has decreased the consumption of energy (Kasulis *et al.*, 1981), and increased the desire for the adoption of renewable energies (Barhoumi *et al.*, 2021; Labay and Kinnear, 1981). Green energy is a form of renewable energy that includes photovoltaic and thermoelectric solar energy, wind, geothermal and biomass energy (Elmustapha *et al.*, 2018; Alam *et al.*, 2014). Researchers reported that greater public environmental consciousness has accelerated the growth of small-scale renewable energy (Alam *et al.*, 2014). According to Frankfurt School – UNEP Collaborating Centre for Climate and Sustainable Energy Finance report, developed countries have heavily invested in small-scale renewable energy projects which included US\$24.1bn investment by Italy and US\$20bn investment by Germany. The report also indicated huge investments by the USA, France, Japan, the UK and Australia in small-scale renewable energy projects (Alam *et al.*, 2014).

Many countries have installed small-scale renewable energy; even installation of small-scale renewable energy has been increased in low populated countries (Solangi *et al.*, 2011). Bilgen *et al.* (2004) reported the surge of renewable energy, and it is expected to increase by 15%–30% at the end of 2030. The study of Güney (2019) included the data of 40 developed and 73 developing countries. According to the findings of the study, renewable energy has a major contribution to the development of both developed and developing countries. The findings of the study also indicated the greater importance of renewable energy than

non-renewable energy on sustainable development. As posited by Waris and Hameed (2020), Pakistan is one of the developing countries that are unable to meet the current energy deficit. The effects of global warming will be more threatening for Pakistan, as it has been listed as the 12th most vulnerable country in the world (Mohsin *et al.*, 2019). To meet the current demand for energy and minimize environmental temperature and combat global warming, Pakistan needs to adopt renewable sources of energies (Arshad *et al.*, 2018). In this regard, green energies such as wind, biomass and solar energies are vital sources that significantly reduce the energy burden on the national grid and help to combat increasing climate change (Menzel and Teubner, 2020). Quinonez-Varela *et al.* (2007) emphasized that domestic consumers have an important role to reduce carbon dioxide emissions by using renewable energy. Further, the energy generated by renewable sources such as wind and solar energies is an important innovation that could effectively use by the domestic consumers of Pakistan.

Many studies have been conducted in the context of developed and developing countries focusing on consumers intention to use smart technologies (Bhutto *et al.*, 2020; Aslam *et al.*, 2020), energy-efficient appliances (Waris and Hameed, 2020), and smart cities (Zavratnik *et al.*, 2020) and energy-efficient technologies (Lee *et al.*, 2020) by the application of extended TPB model. Okwanya *et al.* (2020) highlighted the importance of awareness and income on the communities' usage of renewable sources of energy. In this regard, Sackey *et al.* (2020) emphasized the financial institutions to finance low-income groups for the purchase of solar technology. To the best of researchers' knowledge, studies have not paid much attention to publicity information and green norms which are essential elements of consumers' green technology adoption. Therefore, to fill the gap and understand the domestic consumers' intention to use small-scale solar energy in the context of developing markets, Pakistan, this study has developed an extended theory of planned behavior (TPB) model to predict domestic consumers' intention to use small-scale solar energy. The work of previous scholars has indicated the important role of domestic consumers' to reduce environmental degradation through sustainable energy consumptions (Wang *et al.*, 2018; Aslam *et al.*, 2020). To enrich the body of literature and contribute to the TPB, this study has included public information and green norms as additional constructs of TPB to evaluate domestic consumers' intention to use small-scale solar energy.

2. Literature review

2.1 Theoretical framework and hypotheses development

This study has used an extended TPB model to analyze households' intention to use small-scale solar energy for domestic consumption. Over the years, researchers have used extended models of TPB to assess consumers' intention to use sustainable products in developing countries (Bhutto *et al.*, 2020; Aslam *et al.*, 2020; Hameed *et al.*, 2019; Waris and Hameed, 2020). Ajzen in 1985 provided useful insights regarding the purchase of environmentally friendly products (EFP). The TPB focuses on individual personal assessments of factors affecting intention and behavior, social pressure exerted to perform a particular behavior and attitude to perform a behavior. The TPB is an extended version of the theory of reasoned action (TRA) proposed by Ajzen and Fishbein (1975) that posited that social pressure and attitude lead to intention. TPB included the perceived behavioral control (PBC) constructs in the TRA model and improved the explanatory power of the model. Therefore, this study has used TPB with additional constructs that would predict household consumers' intention to use small-scale solar energy as an alternative source of conventional energy. The extended model of this study has included information publicity and green norms as antecedents of intention to use small-scale solar energy. The additional constructs

to the existing TPB model are vital as it would help to create awareness and affect individual norms to use small-scale solar energy for domestic consumptions.

2.1.1 Attitude. The willpower of the person to perform a particular behavior is influenced by various factors and is referred to as attitude. Attitude is one of the independent variables in the TPB. It explains the strength of the positive or negative behavior towards a phenomenon (Ajzen, 1991; Akhtar *et al.*, 2021; Ha and Janda, 2012). In the context of renewable energy resources, a favorable attitude is the intensity of the person to use alternative energy resources. The behavior to use alternative energy resources is dependent upon the decision-making process of the individual (Lins *et al.*, 2012). Researchers have suggested that the attitude is the significant and consistent predictor of adopting solar energy (Eslami *et al.*, 2021; Zander, 2021; López-Mosquera *et al.*, 2014; Poier, 2021; Kim *et al.*, 2021). It has been claimed by Ajzen (1985) that such behaviors can be performed by people with a positive attitude. Based upon the above discussion, we conclude that attitude has a positive impact on intention to use small-scale solar energy. Thus, it is hypothesized as follows:

H1. Attitude has a positive impact on households' intention to use small-scale solar energy.

2.1.2 Subjective norm. The social pressure on an individual to act or perform a behavior following society's expectations is considered subjective norms (Ajzen, 1991). People high in subjective norms interpret the behavior of others towards an act (Waris *et al.*, 2021). The influence on a person from family, friends and peers, etc. is referred to as a subjective norm (Hameed *et al.*, 2019). The construct subjective norms have also been found to influence the green consumer behavior of the individuals (Hua and Wang, 2019; Kowalska-Pyzalska, 2018; Lopes *et al.*, 2019; Gkargkavouzi *et al.*, 2019). Recently, several studies have proved the power of subjective norms to influence behavior in the adoption of renewable energies (Li *et al.*, 2021; Fang *et al.*, 2021; Lundheim *et al.*, 2021). Therefore, we hypothesize that subjective norms play a positive role in the development of intention to use small-scale solar energy. Thus, it is hypothesized as follows:

H2. A subjective norm has a positive impact on households' intention to use small-scale solar energy.

2.1.3 Perceived behavioral control. Perceived behavior control is the perceived ease of performing a particular behavior (Tan *et al.*, 2017). It is the tendency and ability of an individual to perform behavior under consideration (Klöckner, 2013). Individuals high in PBC tend to demonstrate the same in buying situations (Bagheri *et al.*, 2021; Bui *et al.*, 2021). Perceived behavior control of an individual influences a person favorably to purchase EFP (Hamzah and Tanwir, 2021). Various authors have empirically evaluated and found that perceived behavior control significantly affect pro-environmental behavior (Hameed *et al.*, 2019; Hua and Wang, 2019; Yadav and Pathak, 2017). Particularly, in the context of renewable energies, researchers found that PBC is an important antecedent of adopting renewable energy (Li *et al.*, 2021; Masukujjaman *et al.*, 2021). Based on the above arguments, we infer that PBC has huge importance to influence household intention to use small-scale solar energies. Thus, it is hypothesized as follows:

H3. PBC has a positive impact on households' intention to use small-scale solar energy.

2.2 Green norm

Environmentalists have paid ample attention to psycho-social determinants that influence individual pro-environmental behavior (Klöckner, 2013; Bamberg and Möser, 2007).

The internally motivated consumers establish green norms that depict their demand for the acceptance of green products (Nyborg *et al.*, 2006). Green consumers accept the responsibility of protecting the environment for the benefit of the public (Hameed *et al.*, 2019; Wang *et al.*, 2018). As posited by Dunn *et al.* (2008), people feel satisfied when they participate in charitable work and spend more money on human welfare than spending on themselves. Likewise, people's decisions are based on their relationships with the natural environment (Muradian and Pascual, 2018; Engel *et al.*, 2020). The study of Naderi and Van Steenburg (2018) explored the effects of psychographic variables on millennial green behavior revealed that rational and self-motivated motives lead to sustainable consumption. Concerning the adoption of green electricity, scholars posit that consumers' increasing concern for the environment stimulates the adoption of green electricity (Rowlands *et al.*, 2003; Yang *et al.*, 2016). Based on the above argument, we infer that green norms will positively influence household intention to use small-scale solar energies. Thus, it is hypothesized as follows:

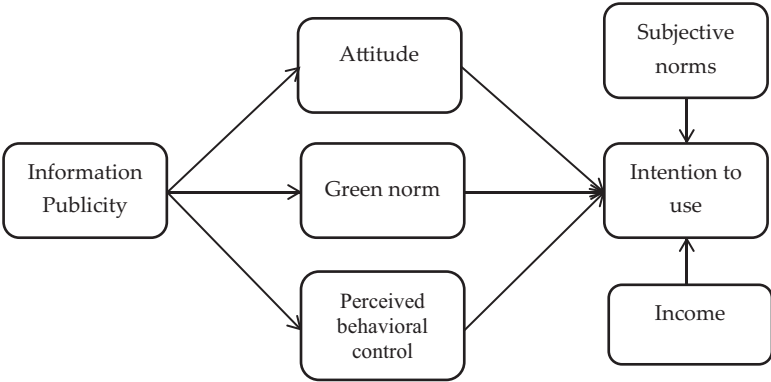
- H4. Green norm has a positive impact on households' intention to use small-scale solar energy.

2.3 Information publicity

Information publicity help to enhance the mechanism and path of consumers' knowledge related to the adoption of green electricity based on the integrated TPB model. Past studies have depicted that information publicity has a significant influence on the behavioral intention of consumers' pro-environmental behavior (Xu *et al.*, 2017). The study of Wang *et al.* (2018) based on the effects of information on residents' e-waste revealed the positive influence of consumers norms on behavioral intention. Steg (2008) emphasized the role of effective information publicity in the conservation of energy. Wang *et al.* (2014) on the determinants of energy saving have confirmed the positive influence of publicity information on residents' attitudes and behavioral intention. Further, other scholars have shown the significance of companies' differentiated information is vital for environmentally concerned consumers (Sammer and Wüstenhagen, 2006), as it helps consumers to understand the sustainable attributes of the products (Waris and Hameed, 2020). Wang *et al.*'s (2018) study revealed the positive effect of publicity information on attitude towards e-waste recycling, but the study findings depict the insignificant influence of publicity information on PBC. Oliver and Lee (2010) found a positive relationship between information seeking and consumers' pro-environmental behavior. A recent study conducted by Zhang *et al.* (2020) revealed that information publicity promotes sustainable behavior and enhances consumer favorable evaluation and acceptance of green electricity (Yang *et al.*, 2016), renewable energy adoption (Szakály *et al.*, 2020; Mahalik *et al.*, 2021). Based on the above arguments regarding the effective role of publicity information in influencing people who intend to use green energy, we hypothesized as follows (Figure 1):

- H5. Information publicity has a positive impact on attitude for intention to use small-scale solar energy.
- H6. Information publicity has a positive impact on PBC for intention to use small-scale solar energy.
- H7. Information publicity has a positive impact green norm of households' intention to use small-scale solar energy.

Figure 1.
Conceptual model



3. Methodology

3.1 Data collection and sampling technique

This study is quantitative in nature and an online survey was conducted to collect the data from the households of seven cities (Karachi, Hyderabad, Islamabad, Lahore, Uthal, Gwadar and Turbat) in Pakistan. The criteria for selecting these cities have a hot climate, diverse ethnic groups and represent the majority population of Pakistan. Households are selected in this study because they are the main consumers of electricity in developing markets, Pakistan. The study of [Cheng *et al.* \(2017\)](#) indicated that household consumers are major contributors of carbon dioxide emission into the environment, therefore, to assess households' intention to use renewable sources of energy such as solar energy adoption is essential to reduce carbon dioxide emission into the environment. Understanding households' energy consumption in the context of developing country Pakistan is even more necessary as domestic consumers account for 85% of energy consumption in Pakistan ([Ali *et al.*, 2019](#)). For the collection of data, a structured questionnaire was designed on Google form and a link of the questionnaire was sent to potential participants of the study through an email and WhatsApp. The researchers collected online data because local governments of different cities had imposed lockdown due to the COVID-19 pandemic. In addition to this, an online method was more feasible to collect the data from remote areas of the country. A total of 836 questionnaires were purposively sent to households living in different parts of these seven cities. Initially, we received only 176 responses from the participants of the study. After sending three reminders with an interval of one week, we were able to gather a total of 374 responses. Before analysis, we assessed multivariate outliers and identified four outliers that were removed from the data.

3.2 Research instrument

A closed-ended questionnaire has been used as an instrument for the data collection. The questionnaire was divided into three sections: the first section contains the cover letter mentioning the purpose of the research and ensuring the confidentiality of the participants. The second and third sections of the questionnaire were related to the demographic details of the participants and constructs used to measure the household intention to use small-scale solar energy, respectively. All the constructs in the study were measured using a five-point Likert scale ranging from strongly disagree = 1 to strongly agree = 5. The study adopted already established scales from previous studies. The attitude consisting of three items was measured by [Taylor and Todd's \(1995\)](#) scale. Subjective norm was consisting of five items

scale adapted from the study of [Hua and Wang \(2019\)](#). PBC was measured by using a five items scale adapted from [Wong et al. \(2012\)](#). Information publicity was measured by five items scale adapted from [Wang et al. \(2018\)](#). Green norms were measured by six items scale adapted from the study of Ozaki and Sevastyanova. (2011). Behavioral intention was measured by four items scale adapted from [Karahanna et al. \(1999\)](#), [Shih and Fang \(2004\)](#) and [Teo and Pok \(2003\)](#) studies. The questionnaire was evaluated by four academic experts from the field of management and marketing. They evaluated the content, layout and language of the instruments. The experts suggested minor changes in the final questionnaire. Before distributing the questionnaire, the pilot test was implemented on 45 households living in Karachi. The results of the pilot test were satisfactory as all items were loaded above 0.60 which is the lowest criterion proposed by researchers ([Hair et al., 2014](#)).

4. Data analysis

4.1 Common method variance

The use of self-reported data and relying on a single source of participants can cause common method variance. According to [Miller and Cardinal \(1994\)](#), anonymous questionnaires are more likely to represent the actual scenario. If a single factor reports for more than half of all variance, substantial common method bias may occur ([Podsakoff et al., 2003](#)). We have used Harman's single factor test to assess the variance explained by a single factor. Since a single factor accounted for only 27.528% variance in data, therefore, we can conclude that common method bias was not an issue in the current data set ([Podsakoff, 2003](#)).

4.2 Profile of the participants

The participants of the study belong to seven different cities of Pakistan. In terms of age, out of 370 participants, 226 were male that constitutes 61.1% of the collected sample. In terms of marital status, 215 participants majority were married that constitutes 58.1% of the collected sample. In terms of qualification, 158 participants hold bachelor's degrees that constitute 42.7% of the collected sample. In terms of monthly income, 157 participants earn around PKR 25000 monthly that constitutes 42.4% of the collected sample. The details of participants have shown in [Table 1](#).

4.3 Reliability and convergent validity

To assess the quality of data, the first condition is to measure internal consistency. We have assessed the internal consistency of the data through Cronbach's alpha values. The values of Cronbach's alpha for all constructs, except publicity information 0.694, are above 0.70,

Gender	Frequency	(%)	Marital status	Frequency	(%)
Male	226	61.1	Unmarried	155	41.9
Female	144	38.9	Married	215	58.1
Qualification			Total	370	100
Matric	1	0.3	Monthly Income		
Intermediate	47	12.7	0PKR to 25000PKR	157	42.4
Bachelor	158	42.7	25001PKR to 50000PKR	63	17.0
Master	152	41.1	50001PKR to 75000PK	33	8.9
PhD	12	3.2	75001PKR to 10000PKR	39	10.5
Total	370	100	100001PKR to 125000PKR	29	7.8
			125001 and above	49	13.2
			Total	370	100

Table 1.
Respondents' profile

meeting the threshold value of ≥ 0.70 . However, [Hair et al. \(2016\)](#) posited that composite reliability (CR) is the better criterion to assess the internal consistency of the data. Therefore, we have also assessed the internal consistency of the data through CR values which depict that all constructs' values exceeded 0.70, confirming data internal consistency. Next, we measured convergent validity which represents the degree to how much a single construct correlates with another construct ([Hair et al., 2016](#)). The convergent validity confirms when the average variance extracted (AVE) of all constructs is equal to or greater than 0.50, and outer loadings are greater than or equal to 0.70 ([Hair et al., 2016](#)). Based on the criteria, the AVEs of all constructs are above 0.50 and CR values are above 0.70 confirming the presence of convergent validity in the data. The details of reliability analysis and convergent validity are shown in [Table 2](#).

4.4 Discriminant validity

Discriminant validity refers to the extent to which a construct is unrelated to another construct in the study ([Alarcón et al., 2015](#)). In this study, we have used three criteria to assess discriminant validity. First, we assessed discriminant validity through [Fornell and Larcker \(1981\)](#) according to which the square of AVEs values must be greater than the corresponding correlations. [Table 3](#) shows the discriminant validity through the [Fornell and Larcker \(1981\)](#) criterion which confirms the presence of discriminant validity. Second, we

Constructs	Items	Loading	Cronbach's alpha	CR	AVE
Attitude	ATD1	0.897	0.880	0.926	0.807
	ATD2	0.904			
	ATD3	0.892			
Behavioral intention	BI1	0.873	0.904	0.933	0.777
	BI2	0.839			
	BI3	0.925			
	BI4	0.885			
Green norm	GN1	0.728	0.819	0.868	0.523
	GN2	0.747			
	GN3	0.682			
	GN4	0.731			
	GN5	0.731			
	GN6	0.719			
Perceived behavioral control	PBC1	0.659	0.805	0.865	0.565
	PBC2	0.639			
	PBC3	0.827			
	PBC4	0.830			
	PBC5	0.780			
Publicity information	PI2	0.732	0.694	0.812	0.522
	PI3	0.614			
	PI4	0.801			
	PI5	0.729			
Subjective norm	SN1	0.795	0.858	0.899	0.642
	SN2	0.824			
	SN3	0.870			
	SN4	0.855			
	SN5	0.642			

Table 2.
Reliability testing
and convergent
validity

Notes: CR = Composite reliability; AVE = Average variance extracted

have used heterotrait–monotrait ratio (HTMT) criterion that requires a construct value must be ≤ 0.85 (Henseler *et al.*, 2015). Table 4 is showing that the values of all constructs are below 0.85. Third, we assessed cross-loading values to confirm the presence of discriminant validity. To confirm discriminant validity, each item of its construct has higher cross-loading values than other constructs as shown in Table 5 (Hair *et al.*, 2014).

4.5 Predictive power of the inner model

The criterion for inner model fit is assessed through two methods: coefficient of determination (R^2) and predictive relevance of the model which is determined through the value of cross-validated redundancy (Q^2) (Hair *et al.*, 2014; Henseler *et al.*, 2009). The value of R^2 represents the variation explained by the exogenous constructs on endogenous constructs. In this study, the value of R^2 for endogenous construct is 57.1% which moderate to high predictive accuracy. Next, we assessed the value of cross-validated redundancy (Q^2) through the blindfolding method. The value of Q^2 above zero indicates the presence of predictive relevance in the model. The results depict the value of Q^2 for the endogenous construct is 43.3% which is considered as substantial predictive relevance for the proposed model of the study.

4.6 Hypotheses testing

The partial least square structural equation modeling (PLS-SEM) has been used to test the proposed hypotheses using 5000 bootstrapping sampling method (Haenlein and Kaplan, 2004). The results of the hypotheses revealed that PBC has the strongest impact on intention to use small scale solar energy ($\beta = 0.349$; $t = 8.277$; $p < 0.05$), followed by attitude ($\beta = 0.295$; $t = 5.921$; $p < 0.05$), green norm ($\beta = 0.187$; $t = 5.062$; $p < 0.05$) and subjective norm ($\beta = 0.163$; $t = 2.990$; $p < 0.05$). In addition, the results revealed the publicity information has strongest impact on green norm ($\beta = 0.384$; $t = 8.904$; $p < 0.05$), followed by PBC ($\beta = 0.252$; $t = 4.483$; $p < 0.05$) and attitude ($\beta = 0.255$; $t = 4.461$; $p < 0.05$). The results

Latent variables	1	2	3	4	5	6
Attitude	<i>0.898</i>					
Behavioral intention	0.558	<i>0.881</i>				
Green norm	0.244	0.347	<i>0.723</i>			
Perceived behavioral control	0.288	0.578	0.125	<i>0.752</i>		
Publicity information	0.255	0.352	0.383	0.252	<i>0.722</i>	
Subjective norm	0.602	0.568	0.216	0.493	0.253	<i>0.801</i>

Table 3.

Notes: The italic diagonal values refer to the square root of the AVE of each construct. All correlations are statistically significant ($p < 0.01$)

Discriminant validity
analysis

Latent variables	1	2	3	4	5	6
Attitude						
Behavioral intention	0.622					
Green norm	0.283	0.390				
Perceived behavioral control	0.331	0.664	0.225			
Publicity information	0.324	0.426	0.496	0.313		
Subjective norm	0.674	0.638	0.258	0.599	0.330	

Table 4.

Heterotrait–
monotrait ratio
(HTMT) results

Table 5.
Cross loadings

	ATD	BI	EC	GN	PBC	PI	SN
ATD1	0.897	0.467	0.282	0.245	0.239	0.218	0.494
ATD2	0.904	0.499	0.320	0.206	0.303	0.256	0.579
ATD3	0.892	0.534	0.291	0.255	0.236	0.212	0.547
BI1	0.479	0.873	0.404	0.363	0.479	0.314	0.470
BI2	0.430	0.839	0.290	0.301	0.460	0.268	0.461
BI3	0.547	0.925	0.392	0.382	0.528	0.345	0.557
BI4	0.502	0.885	0.339	0.315	0.570	0.311	0.510
GN1	0.208	0.302	0.396	0.728	0.084	0.327	0.203
GN2	0.277	0.240	0.478	0.747	-0.050	0.273	0.163
GN4	0.122	0.269	0.227	0.682	0.273	0.231	0.176
GN5	0.141	0.293	0.342	0.731	0.279	0.230	0.170
GN6	0.141	0.256	0.316	0.731	0.305	0.252	0.155
PBC1	0.182	0.310	0.028	0.110	0.659	0.125	0.324
PBC2	0.339	0.489	0.216	0.135	0.639	0.188	0.424
PBC3	0.287	0.482	0.171	0.192	0.827	0.220	0.452
PBC4	0.122	0.432	0.070	0.212	0.830	0.217	0.326
PBC5	0.116	0.413	0.142	0.207	0.780	0.178	0.297
PI2	0.171	0.227	0.309	0.228	0.124	0.732	0.177
PI3	0.175	0.124	0.369	0.257	0.088	0.614	0.153
PI4	0.176	0.263	0.269	0.302	0.176	0.801	0.210
PI5	0.210	0.362	0.256	0.295	0.301	0.729	0.190
SN1	0.589	0.520	0.359	0.218	0.360	0.227	0.795
SN2	0.475	0.450	0.310	0.265	0.419	0.227	0.824
SN3	0.512	0.466	0.251	0.196	0.380	0.139	0.870
SN4	0.530	0.471	0.287	0.179	0.390	0.205	0.855
SN5	0.239	0.343	0.169	0.113	0.466	0.227	0.642

for the study depict that PBC, attitude, green norm and publicity information are important constructs that appeared to impact households' intention to use small scale solar energy. Further, we have analyzed the effect of income on household behavioral intention to use solar energy. The results of the study depict that income has significant impact of households' intention to use solar energy ($\beta = 0.153$; $t = 4.429$; $p < 0.05$) (Table 6) (Figure 2).

5. Discussion and conclusion

The rising temperature of the environment due to carbon dioxide emission has threatened environmental sustainability. Therefore, the world has focused on sustainable resources to

Table 6.
Results of the
structural model

Hypotheses	Beta	<i>p</i> -values	<i>t</i> -values	Decision	Effect
ATD → INT	0.295	0.000	5.921	Supported	Large
SN → INT	0.163	0.003	2.990	Supported	Small
PBC → INT	0.349	0.000	8.277	Supported	Small
GN → INT	0.187	0.000	5.062	Supported	Large
PI → ATD	0.255	0.000	4.461	Supported	Small
PI → PBC	0.252	0.000	4.483	Supported	Small
PI → GN	0.384	0.000	8.904	Supported	Large
Income → INT	0.153	0.000	4.429	Supported	Small

Notes: Beta = Standardized coefficient path, SE = Standard error, Significant at ($p < 0.05$)

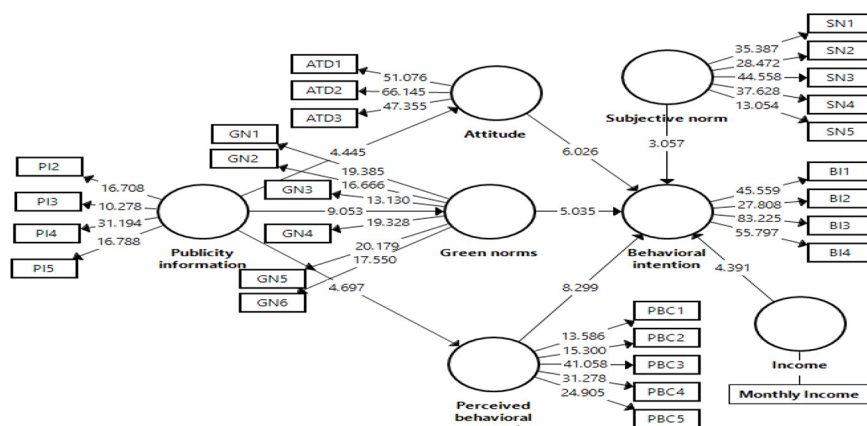


Figure 2.
Result of structural
model

combat environmental issues. In the past two decades, the world has seen a substantial climate change that resulted in environmental pollution, floods and earthquakes. This study aims to predict households' intention to use small-scale solar energy for domestic consumption in a developing country, Pakistan. Pakistan is among the 12th most vulnerable countries in terms of climate change (Mohsin *et al.*, 2019). The energy consumption from non-renewable sources of energy will further severe climate change in the country as more than 46% of the domestic population consumes energy through non-renewable resources of energy such as fuel (Ali *et al.*, 2019). Therefore, the use of renewable energy sources has become inevitable to deal climate the ever-increasing temperature of the environment and combat climate change. This study attempts to understand households' usage of small-scale solar energy in Pakistan. This study is based on an extended TPB model which includes potentially significant constructs that play an important to predict households' intention to use small-scale solar energy. Publicity information and green norm have been added as an additional construct in the extended TPB model. Researchers argued that publicity information and green norm are essential elements in predicting consumers' intention to use green energy (Yang *et al.*, 2016; Fornara *et al.*, 2016; Ek and Söderholm, 2008). The results of the study revealed that household attitude has a significant impact on intention to use small scale solar energy which is consistent with the findings of Eslami *et al.* (2021) and Zander (2021), the authors argued that small scale solar energies are affordable and self-dependent that develop a positive inclination towards the adoption of technology. The positive impact of the subjective norm has also been verified signifying that peers and friends, and social community influence the adoption of small-scale solar energy which matches with the findings of previous researchers (Li *et al.*, 2021; Lopes *et al.*, 2019). In terms of the positive impact of PBC on intention to use small-scale solar energy, the study found that it is a significant factor of household intention to use solar energy. The findings are consistent with past studies where researchers found that PBC is an important antecedent of pro-environmental behavior, especially in the context of sustainable energy consumptions (Bhutto *et al.*, 2020; Waris and Hameed, 2020; Aslam *et al.*, 2020). In terms of the positive impact of the green norm on intention, the results of the study found that the green norm is a strong predictor of households' intention to use small-scale solar energy. In past, researchers argued that Pakistani have a deep emotional attachment with the religion therefore, they are philanthropists by nature and want to contribute more for the betterment of society and the environment (Aslam *et al.*, 2020; Hameed *et al.*, 2019). Green norms are established beliefs of

people to follow green values and conserve the environment through responsible behavior. Additionally, the results of this study support that publicity information is an important antecedent of TPB constructs (attitude, green norms and PBC) that explain that it is the first and crucial step in establishing sustainable norms and attracting consumers towards the adoption of small scale solar energy. These findings match with the past studies where authors argued that information has a crucial impact on decision-making regarding sustainable behaviors (Wang *et al.*, 2018; Waris and Hameed, 2020; Alam *et al.*, 2014; Mirza *et al.*, 2009). The results further indicate that publicity information has the strongest impact on green norms which signifies the importance of information accessibility in green electricity adoption.

6. Implications

6.1 Theoretical implications

This study has contributed to the TPB and renewable energy consumption. The current study has added two additional constructs into the TPB to better explain the renewable energy consumption of households in a developing country, Pakistan. Publicity information has been taken as an antecedent of TPB constructs and explained that it has a substantial impact on PBC and attitude. This signifies the importance of information sharing and public capacity to accept solar energy as a sustainable means of consumption. Additionally, this study has included a novel construct of the green norm. Some scholars have prior studied the green norm in the green energy perspective (Nyborg *et al.*, 2006; Rowlands *et al.*, 2003), but the green norm has not been integrated into the TPB model. In addition, scholars have not paid much attention to publicity and the green norms relationship. This study has developed and proved that publicity information leads to green norms which eventually affect households' adoption of small-scale solar energy. Past studies have, however, studied moral norms and personal norms and integrated both with TPB. The Green norm is an important construct in the consumption of adoption of green products (Rowlands *et al.*, 2003). Therefore, this study has included green norm into the TPB model.

6.2 Policy implications

Despite several studies regarding sustainable energy consumption, very few studies have focused on households' intention of using small-scale solar energy, especially from the perspective of developing countries. Past studies have paid ample attention to the energy efficiency techniques and smart technologies to conserve energy and protect the natural environment (Waris and Hameed, 2020; Aslam *et al.*, 2020; Bhutto *et al.*, 2020). From the policy perspective, understanding new technology drivers and people's response to technology is essential (Ferroukhi *et al.*, 2013; Kolloch and Reck, 2017). This study is novel and practically contributes to the literature by understanding households' intention to adopt solar technologies that are viable means to conserve conventional energy and preserve the environment through less emission of carbon dioxide. In addition to this, understanding green norm of households is imperative in a developing country Pakistan where climate risk is high. This would help marketers and practitioners to design and introduce new and more efficient renewable technologies that maintain environmental sustainability. In light of the current study findings, several implications emerge regarding the adoption of solar technology by households. First, the study highlighted the importance of publicity information in developing a positive attitude and establishing green norms. Second, publicity information increases personal self-efficacy to adopt and use solar technology as an alternative option because of the viability of green technology. In addition to this, the green norm which is an important aspect of household adoption of green technology and

sustainable consumption was developed by publicity information. The findings revealed that people who have developed green norms are more likely to adopt small-scale solar technology. Third, the adoption of small-scale solar energy at the mass level would reduce the burden on the national grid and eventually help to reduce carbon dioxide emissions into the environment. In addition to this, the reduced consumption of conventional energy would reduce fuel consumption and financially stabilize the developing economy. Therefore, governments have to introduce schemes and awareness programs that disseminate information about the viability of small-scale solar energy and support the households through microfinancing facilities to adopt small-scale solar energy and improve environmental sustainability.

7. Limitations and recommendations

Although this study has covered several new aspects and contributed to the TPB and renewable energy literature, there are some limitations to this study. First, the study has taken solar energy as renewable energy means for households consumption, there are many other renewable energy options available and being used in several countries such as wind energy, biomass energy, hydro, tidal. Second, this study has quantitative research design and online survey conducted to collect the data of the households. Future researchers can conduct qualitative studies and use in-depth interviews to explore more that may be enablers and disablers in the adoption of renewable energies. In addition to this, future studies can include several other variables that are essential in the context of pro-environmental behavior such as altruism, personal norms and moral norms. Third, future studies can be conducted by using the triple helix model to understand the support from these three pillars in the adoption and use of renewable technologies.

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Further reading

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